

Understanding the effect of age on a soccer player's current market value

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Abstract

The aim of this project is to answer the big question of what the effect of age on a soccer player's current market value is. With the spark of young talent lately in soccer and record-breaking transfers for senior players as well, the age factor has remained a crucial point in deciding player's market value. Using a dataset consisting of players' statistics and performance metrics, this study aims to decipher exactly how much a player's current market value is affected from their age. The study will incorporate several different regression analysis techniques and see if the values are statistically significant and are a good estimate for the real-world scenario. This project gives us the idea of the importance of analysis in sports and will make it easier for professionals to make informed decisions, telling us that a player's value is never random and can be explained using the right methods.

Introduction

The biggest sport in the world is undoubtedly soccer. The 2022 world cup final of France vs Argentina recorded a total of 1.5 billion viewers highlighting how famous the sport is. With such sports, rising talent is everywhere. With new emerging players like Lamine Yamal, who at 17 years old has a market value of \$196 million and senior solidified players like Mohammad Salah, who at 32 years old still has a market value of \$65 million. A player's market value is a crucial indicator of their perceived worth and potential contribution to a team. Clubs, agents, and analysts closely monitor market values to make informed decisions about transfers, contracts, and squad investments. While numerous factors influence a player's value—such as performance metrics, position, league strength, and injury history—one of the most debated determinants is age. Age plays a dual role in shaping a player's worth. As stated by Margareta et.al, “the useful life of a football player decreases over time, giving rise to the assumption that the older the

player, the less benefit is left”. (Maragareta et.al) Understanding how age influences a soccer player’s market value is essential for clubs aiming to maximize their return on investment and for players negotiating contracts at different stages of their careers. There have been several moments in soccer which go against the norm that as a players age, they lose market value. Barbuscak states that his reports suggest “transfer fees started declining after players reached 25 years of age.” (Barbuscak) However, everyone saw Neymar’s record-breaking transfer value of \$240 million from Barcelona to PSG. This was after he was 25 and he his market value stayed high till he was 30. Hence, the goal of this project is to answer the question what is the effect of age on a player’s current market value? This study employs an empirical approach, utilizing data from a range of professional leagues to analyze the effect of age on market value. By applying statistical models learnt such as OLS regression, probit regression, and non-linear regression to player valuation data, the question can be answered well. We can also how close the estimate is to the real-world scenario and can infer if players are being over or under-valued in the present day. The findings of this research have implications for multiple stakeholders in the soccer industry. Clubs can refine their transfer strategies to optimize squad investments, agents can use insights to better position their clients in negotiations, and analysts can enhance forecasting models for player valuation. Furthermore, understanding the role of age in market value contributes to broader discussions in labor economics and sports management, offering valuable lessons applicable beyond soccer. Findings from Pérez-González et al. (2020) suggest that the impact of age on market value is further influenced by market dynamics, as younger players with potential may be overvalued at the expense of experienced professionals. This relates exactly to the main question being answered as this study in fact shows external factors on top of the player’s age seem to influence player values. Another study by Brito (2020) concluded that

“forwards peaked at 25 in terms of matches played and at 28 in terms of goals scored”. The two important factors which determine a player’s market value. However, some limitations with the existing study include that they don’t consider data from multiple different leagues across the world. Brito’s study focuses mainly on Bundesliga whilst Perez focuses mainly on the premier league. While these are major leagues in the world of soccer, we also must take into the other leagues in Europe and in Asia as that’s a growing market for soccer as well. With the rise of players transferring to the Saudi league, player values do get inflated with ridiculous deals circling through. Kalen et.al (2019) concluded by looking at champion league data, which consists of the biggest clubs in Europe that “the average forward and defender peaks at 25 and 27, respectively. For midfielders, the peak age occurs in the 25–27 age range.” Kalen’s study indicates that after 27 years, a player’s value starts to decline. This also makes sense due to the different demands that each position requires. These results provide great motivation for the project because they do indicate a major trend for age and player values and prove that using age is a great factor to determine a player’s value for clubs when making decisions. Due to the increase of matches in modern soccer matches, studies do indicate that older players tend to perform worse than the younger talent, however that does not discount their technical acumen as experience also plays a major role. As Margareta et.al’s study (2022) indicates that age often has a negative effect, suggesting that clubs may undervalue older players despite their experience and tactical acumen. We see examples of older players still dominating the game. Luka Modric for example, a 39-year-old midfielder plays for Real Madrid and is often a starting player still in his top form. He even gets called up to play internationally for Croatia. His market value now is around \$8 million, but 5-years ago when he was 34 years old, he was still valued at a staggering \$45 million value. This indicates that technical abilities also play a major role on top of age.

Knowledge of when players are in their optimal age, therefore, has substantial value for the soccer industry. However, another major motivation for this study rises from the fact that no studies have examined the aging trend of male elite soccer players in recent decades. Conversely, aging trends in different individual and team sports like baseball or tennis have been previously investigated, suggesting a marked increase in the age of peak performance of elite athletes during the last two decades, probably due to major advances in sport science and technology. Hence, knowing the effect of a soccer player's age on their current market value is of great importance to the industry as it sheds light on the dynamics of professional sports economics and allows clubs to choose the best talent to add to their teams and know the type of players to target or sell in the transfer market to maximize their net profits.

Dataset

The dataset features player performance metrics from the 2021-2022 and 2022-2023 seasons. The player data includes basic information such as age, height, playing position, as well as professional statistics. The data was taken from Kaggle, an open dataset website. The authors who posted the dataset compiled the data from Transfrmarket, a well-known website that displays player information. Additionally, variables named "goals," "assists," "yellow cards," "second yellow cards," "red cards," "goals conceded," and "clean sheets" in the dataset were transformed to a per 90 minutes basis. This was done by dividing each statistical value by 90 minutes, as that is the total duration of a normal soccer match. This makes the data more enhanced for evaluation displays a more detailed analysis of the player's performance on a game-by-game basis. The initial dataset included 10,584 observations as it collects the records of players from 11 European leagues, 4 American leagues, 1 African league, and 4 Asian leagues. There are 22 columns indicating the 22 different variables provided of each player. However,

after cleaning the data in the software STATA, the sample size drops to 9,525 observations with the same 22 columns. This was done by removing the goalkeepers recorded since we are using goals scored as a control variable. The number of goals from goalkeepers are always going to be low. Hence, we only look at outfield players as that gives us a better general estimate than also using goalkeeper data. As there are many variables given, the ones that will be used for this project, which seemed to have a high correlation with age and player value are:

Variable	Description
age	The age of the player in years. This is our main parameter of interest.
current_value	The current value of a player in euros. This will be our dependent variable.
goals	The number of goals scored per 90 minutes.
assists	The number of assists per 90 minutes.
height	The height of the player in cm
award	The number of awards the player has received in his lifetime.
winger	This is a dummy variable where 1 is if the player is a winger and 2 is if the player isn't a winger.

Table 1: List of variables and their descriptions

Initial Analysis

Variable	Obs	Mean	Std. dev.	Min	Max
current_value	9,525	3877438	9477191	0	1.80e+08
age	9,525	25.85822	4.626694	15	43
goals	9,525	.1417014	.2457172	0	11.25
assists	9,525	.0982003	.148658	0	4
height	9,525	180.2614	6.600857	156	206
winger	9,525	.3471916	.4761023	0	1

Figure 1: Summary statistics of selected variables

Upon conducting some initial analysis of the data from the summary statistics, we see that the mean value of a player's current value is €23,877,438. Not a majorly high value as this tells us initially that the players are valued at a reasonable rate. However, the standard deviation is also

high which is worth noting of €9,477,191, indicating a high deviation for the player values. The mean age of the dataset is around 25 years.

Upon doing an initial scatter plot of the current player value and age, we get the following:

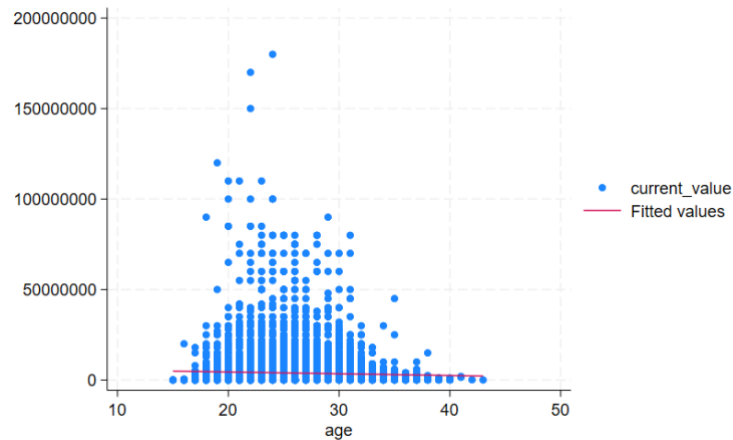


Figure 2: Scatter plot with regression line between age and current value

As we can see from the plot there is a weak but negative trend between the age and current value from the linear regression fit meaning that as age increases, the player's value decreases.

However, we see it doesn't capture all the data suggesting that non-linear terms may be needed and that other factors also play a hand in determining player value. There are also only a handful of players above €100 million who are likely exceptional talents, that too within the age range of 20-25 years old.

Results

To answer our main question, we first use a simple OLS linear regression. An OLS, ordinary least squares, regression is conducted by minimizing the sum of squared differences between the observed and predicted values, helping to model the relationship between the variables. Our first regression estimation equation is: $\widehat{current_value} = \widehat{\beta}_0 + \widehat{\beta}_1 age$. We are also assuming the least square assumptions that there is zero conditional mean, the X and the Y are independent, large

outliers are unlikely, and there is no perfect collinearity for all our regression analysis throughout. This means that the expected value of the error term is 0 as we assume our X values to be independent and we also are saying that no two regressors are linear components of each other. By simply regressing *current_value* on *age*, we get the value of our $\widehat{\beta}_1$ as -98007.46. This tells us that as *age* increases by one unit, the *current_value* of the player decreases by €98,007.46. However, as we saw in the scatter plot, there isn't a constant linear decline in player values as *age* increases. In fact, in real life, young players tend to increase in their value as they age until approximately 25 and then slowly begin to decline. Let's conduct a single restriction hypothesis test to see if the value is significant. We are going to make our $H_0: \widehat{\beta}_1 = 0$ and our $H_A: \widehat{\beta}_1 \neq 0$. The purpose of this test is to assess whether there's a statistically significant relationship between *age* and *current_value*. From our hypothesis test, we get a t-value of -6.08. Taking a 95% confidence level and the significance value of 1.96, our value of $|6.08| > 1.96$. This indicates that we reject the null hypothesis and conclude that the value is indeed statistically significant as the effect of *age* on a player's value is likely not 0. However, a simple regression like this often assumes that all players lose their value at the same rate, which is not an accurate measure. Hence, we can also use interaction terms and non-linear regressions. For our first alteration, we can use the following estimation equation:

$$\widehat{current_value} = \widehat{\beta}_0 + \widehat{\beta}_1 age + \widehat{\beta}_2 age^2 + \widehat{\beta}_3 height * goals$$

We implement the interaction terms between *age* as we saw that there is likely a non-linear relationship between *age* and *current value*. Hence, we add a quadratic term. Additionally, we also added an interaction term between the player's height and goals scored because we also want to add to the regression as taller players tend to score more goals and that does influence their current market value and taller players also may tend to hold their value for longer due to

their physical ability. Hence, including them is important to also avoid omitted variable bias, which occurs when a model fails to include relevant variables. Here, to calculate the effect of age on the `current_value`, we must take the derivative as we also have the interaction term between the ages. This is because when a quadratic term is included, the effect of age is not constant—it changes depending on age itself. Hence, we calculate the partial derivative of the equation with respect to age giving us: $\frac{d\widehat{current_value}}{dage} = \widehat{\beta}_1 + 2\widehat{\beta}_2age$. Our $\widehat{\beta}_1$ value is 2532913 and our $\widehat{\beta}_2$ is -49423.66. Thus, our equation becomes $2532913 + 2 * (-49423.66)age$. This equation tells us how a one-year increase in age affects a player's market value, depending on their current age. We see at early ages, the intercept dominates, meaning a player's value increases, but as the player ages their value tends to drop. We get on average using the partial derivative that as a person's age increases by one year, their value drops by €23,102.37. A big difference and a more realistic answer for the soccer world. We can also add more terms to enhance our regression and capture the effect in a better manner. Our next regression equation comes to: $\widehat{current_value} = \widehat{\beta}_0 + \widehat{\beta}_1age + \widehat{\beta}_2age^2 + \widehat{\beta}_3age * goals + \widehat{\beta}_4age * award$. The results of this regression can be seen by the following table:

	(1)
current_va~e	

age	3538597.5*** (0.000)
c.age#c.age	-73883.2*** (0.000)
c.age#c.go~s	118993.3*** (0.000)
c.age#c.aw~d	25746.4*** (0.000)
_cons	-38551897.5***

	(0.000)

N	9525
R-sq	0.122
adj. R-sq	0.122

p-values in parentheses	
* p<0.05, ** p<0.01, *** p<0.001	

Table 3: Output of the listed above multiple regression

The table tells us about the coefficients for the terms that we regressed upon. In this case, to calculate the effect of age on the current value, we must do another partial derivative with respect

to age. This time we get the equation: $\frac{d\widehat{current_value}}{dage} = \widehat{\beta}_1 + 2\widehat{\beta}_2age + \widehat{\beta}_3goal + \widehat{\beta}_4award$.

Inputting our coefficients, we get: $\frac{d\widehat{current_value}}{dage} = 3538597.5 + 2 * (-73883.2)age +$

$118993.3goal + 25746.4award$. By using the margins command in the software STATA, for this regression on average we get that as increases by one unit, the value decreases by €-

214,725.9. This value is different because here we also incorporate for the interaction between the age and awards. If a player receives an award their value does seem to increase, but it shows that age has a stronger negative impact on player value than previously estimated. To ensure an even clearer result, we can conduct another form of non-linear regression this time our equation

being: $\widehat{current_value} = \widehat{\beta}_0 + \widehat{\beta}_1age + \widehat{\beta}_2age^2 + \widehat{\beta}_3age * goals + \widehat{\beta}_4age * award + \widehat{\beta}_5age * assists + \widehat{\beta}_6age * winger$

From this to accurately calculate the effect of age on the current value, we conduct the partial

derivative with respect to age again and get: $\frac{d\widehat{current_value}}{dage} = \widehat{\beta}_1 + 2\widehat{\beta}_2age + \widehat{\beta}_3goal +$

$\widehat{\beta}_4award + \widehat{\beta}_5assists + \widehat{\beta}_6winger$

Using the margins command on STATA again this time we find that on average that as a player's age increases by one year, their value decreases by €212,004.2. We see yet another high value indicating a strong relation between the variables. This makes sense as well because as players age, they start to score less goals and make less assists and thus would win less awards. All this in total negatively affects their player value. To see the importance of this value, we can conduct a multiple restriction hypothesis test also known as an F-test. Let's make our $H_0: \hat{\beta}_1 = \hat{\beta}_2 = \hat{\beta}_3 = \hat{\beta}_4 = \hat{\beta}_5 = \hat{\beta}_6 = 0$ and our H_A : at least one of them is not equal to 0. From our F-test we get a value of 60.60. Since we have 6 regressors here that we are testing we are going to use the critical value is going to be 2.10. Since $60.60 > 2.10$, we reject the null hypothesis and we say that there is not a zero-effect of age on current value, meaning that age does influence our players' current market value. We say that Age and its interactions terms significantly affect the player value.

However, one thing worth noting is that our variable age can still be endogenous. Endogeneity occurs when our primary variable of interest is correlated with the error term u . This violates the OLS assumption of X being exogenous that we assume and leads to biased and inconsistent estimates. This is because we can also argue that a highly valued young player like Mbappé or Lamine Yamal may receive better contracts and stay in the market longer as he started his development at an early age and still is growing and increasing his value as he gets older. This makes age an endogenous variable as we see a reverse causality approach meaning that instead of seeing the effect of age on the market value, market value also influences career longevity which affects the overall ages of players and their values too. Additionally, something like the player's starting contract can be influenced with age and affect their current market value. To remove this endogeneity issue we need a good instrumental variable. An observed variable Z is

an instrumental variable (IV) when its covariance with X is not 0. This means $cov(Z, X) \neq 0$.

The second condition is that the covariance between Z and the error term u is 0. This means $cov(Z, u) = 0$. In the case of this study, the IV must be correlated with age but must not be correlated with the error term (u), meaning it should affect market value only through age and no other way. One such potential variable that we can use is the player's debut age in professional soccer. Players who enter professional soccer earlier might have different career trajectories, but their debut age is not directly determined by their current market value, making it a potential IV. Additionally, a lower debut age means the player starts playing professionally earlier, increasing their likelihood of reaching an older age and continue playing due to the early opportunities given, making them reach an older age in the dataset.

Another way to determine the effect of age on the current value is through a regression technique called probit regression. This regression models the probability of a binary outcome based on one or more predictor variables, using a cumulative standard normal distribution as its link function.

In our case, we can make our dependent variable current value binary by in effect calculating the estimated effect of age on the probability that the value of the player is greater than the median.

This is a significant metric because it allows us to quantify the likelihood of a player being highly valued based on age, which is useful for talent evaluation for clubs in the real world.

Additionally, by focusing on whether a player is above or below the median, we reduce the impact of measurement errors in absolute valuations. Our estimated regression equation for

probit regression model will look like: $\Pr(\widehat{medianvalue} = 1 | X_1, X_2, X_3, X_4, X_5, X_6) = \Phi(\widehat{\beta}_0 + \widehat{\beta}_1 age + \widehat{\beta}_2 age^2 + \widehat{\beta}_3 age * goals + \widehat{\beta}_4 age * award + \widehat{\beta}_5 age * assists + \widehat{\beta}_6 age * winger$

Yet again to interpret the actual effect of age on the probability we must take the partial derivative with respect to age and we get that on average that as a player's age increases, the

probability that their current market value is greater than the median of €900,000 increases by 0.0002766 or 0.02766%. This value is essential because it accounts for the young talent age theory that we discussed indicating that as younger players age, their value increases. Because as we see the marginal effect when age is 20, 25, and when age is 30, we get the respective probability values of .0718272, .0074748, and -.0555274. This indicates that as the player ages over 30 the probability that their current market value is higher than the median value decreases by 5.55% whilst if a 25 year old person ages one more year, their probability of being higher than the median is estimated to increase by 0.74748%, indicating that the ages of players have a major effect of current market value and that not all players' value changes the same way as they age. Whilst if a 20-year-old player would age one year their probability value is likely to increase by 7.18% meaning that there is a higher change a younger player has a higher salary.

Conclusion

Overall, we see that there is an overall strongly negative effect of age on a player's current value. However, it is important to note from our regression analysis that it is unrealistic to assume that players' market value decrease at a constant rate as they age. In fact, we found that younger players below 25 tend to increase their market value as they age and then it slowly starts to decrease as we saw with the probit regression model showing the probability that the estimated value being greater than the median of €900,000 is quite high for younger players and is quite low for older players. Additionally, we see that it likely there is a nonlinear relation between age and the player value. However, contrary to some studies which discounted age as a major factor for determining a player's value, this study concludes that age in fact is a strong indicator and has a high impact on the current value of a soccer player, further indicating that as clubs decide who to target in the transfer market, age should be an important consideration along with the

player's statistics in other abilities like their goals scored, assists made, and awards received in their respective career.

References

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A. Appendix

A.1 Tables and Figures

```
. regress current_value age,vce(robust)
```

```
Linear regression               Number of obs   =      9,525
                               F(1, 9523)         =      36.95
                               Prob > F           =      0.0000
                               R-squared           =      0.0023
                               Root MSE        =      9.5e+06
```

current_value	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	-98007.46	16123.63	-6.08	0.000	-129613.2	-66401.72
_cons	6411736	464099.6	13.82	0.000	5502002	7321470

Figure 3: OLS regression of output of current_value against age

Linear regression

Number of obs

F(3, 9521)

Prob > F

R-squared

Root MSE

=

=

=

=

=

9,525

120.10

0.0000

0.0313

9.3e+06

current_vare	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	2532913	166405.7	15.22	0.000	2206723	2859104
c.age#c.age	-49423.66	3051.812	-16.19	0.000	-55405.86	-43441.45
c.height# c.goals	25156.87	6845.495	3.67	0.000	11738.24	38575.5
_cons	-2.82e+07	2224795	-12.66	0.000	-3.25e+07	-2.38e+07

Figure 4: regression output for first multiple regression conducted in analysis

Linear regression		Number of obs	=	9,525
		F(4, 9520)	=	87.96
		Prob > F	=	0.0000
		R-squared	=	0.1219
		Root MSE	=	8.9e+06

current_value	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	3538597	209977	16.85	0.000	3126998	3950197
c.age#c.age	-73883.23	4179.695	-17.68	0.000	-82076.33	-65690.14
c.age#c.goals	118993.3	29467.07	4.04	0.000	61231.58	176755.1
c.age#c.award	25746.44	2827.982	9.10	0.000	20202.99	31289.89
_cons	-3.86e+07	2630654	-14.65	0.000	-4.37e+07	-3.34e+07

Figure 5: Second multiple regression output conducted in analysis

Linear regression		Number of obs	=	9,525
		F(6, 9518)	=	60.60
		Prob > F	=	0.0000
		R-squared	=	0.1283
		Root MSE	=	8.9e+06

current_value	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	3507100	209261.7	16.76	0.000	3096902	3917297
c.age#c.age	-73369.8	4160.609	-17.63	0.000	-81525.48	-65214.12
c.age#c.goals	95206.32	26016.41	3.66	0.000	44208.6	146204
c.age#c.award	25116.65	2778.166	9.04	0.000	19670.85	30562.45
c.age#c.assists	202094.8	35449.35	5.70	0.000	132606.5	271583
c.age#c.winger	-21792.64	6880.796	-3.17	0.002	-35280.47	-8304.812
_cons	-3.83e+07	2622784	-14.60	0.000	-4.34e+07	-3.31e+07

Figure 6: Third multiple regression output conducted in analysis

```
. test age c.age#c.age c.age#c.goals c.age#c.award c.age#c.assists c.age#c.winger
> r

( 1) age = 0
( 2) c.age#c.age = 0
( 3) c.age#c.goals = 0
( 4) c.age#c.award = 0
( 5) c.age#c.assists = 0
( 6) c.age#c.winger = 0

F( 6, 9518) = 60.60
Prob > F = 0.0000
```

Figure 7: Multi-restriction hypothesis test output from STATA

	(1)	(2)	(3)
	current_va~e	binary_Y	binary_Y
main			
age	3538597.5*** (0.000)	0.864*** (0.000)	0.868*** (0.000)
c.age#c.age	-73883.2*** (0.000)	-0.0171*** (0.000)	-0.0171*** (0.000)
c.age#c.go~s	118993.3*** (0.000)	0.0131*** (0.000)	0.0128** (0.003)
c.age#c.aw~d	25746.4*** (0.000)	0.00433*** (0.000)	0.00433*** (0.000)
c.age#c.as~s		0.0296*** (0.000)	0.0304*** (0.000)
c.age#c.wi~r			-0.00197 (0.086)
_cons	-38551897.5*** (0.000)	-10.95*** (0.000)	-10.99*** (0.000)
N	9525	9525	9525
R-sq	0.122		
adj. R-sq	0.122		
p-values in parentheses			
* p<0.05, ** p<0.01, *** p<0.001			

Table 3: Probit regression table output